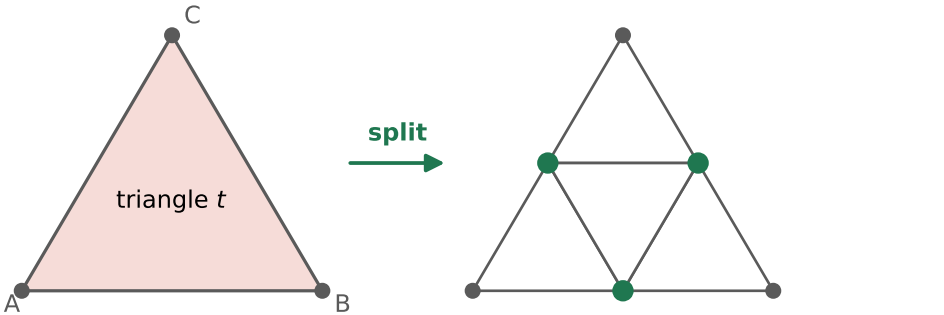
(a) priced split: does this triangle deserve 30 more bytes?

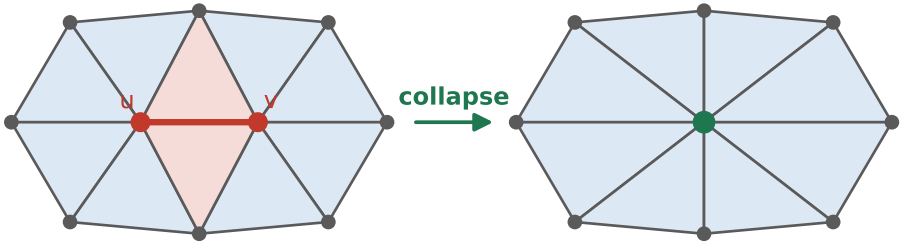


3 new vertices (edge midpoints), 1 parent -> 4 children

Stage 1, every 500 it until 10k: split t iff $\Delta D_t/\Delta R_t > \lambda$

ΔD_t: in each training view, sum the L1 error of the pixels
whose front triangle is t, divide by image Mpx, average over views
e.g. 40 px x 0.05 = 2.0 ; / 1.6 Mpx = 1.25 ; visible in 30 % of views -> 0.375
$\Delta R_t = 3B_a + 6B_f$ = 3 x 6 B + 6 x 2 B = 30 B
(3 new vertices; after the Delaunay rebuild each vertex brings ~2 faces)
ratio = 0.375 / 30 = 0.0125
λ = 6.2e-3 : 0.0125 > λ -> split
λ = 1.85e-2: 0.0125 < λ -> keep (growth stops here earlier)

(b) lambda-prune: is this vertex worth its 10 bytes?



10 faces in the two rings
(red = the 2 faces on edge uv)

merged vertex (green) at the midpoint

8 faces: -1 vertex, -2 faces

Stage 2, at 12k / 14k / 16k / 18k: collapse (u, v) iff $\Delta D_{uv}/\Delta R_{uv} < \lambda$

ΔD_{uv} = sum of ΔD_f over all faces f touching u or v
(same per-view, per-Mpx units as the split), e.g. 10 faces x 0.004 = 0.04
$\Delta R_{uv} = B_a + n_{uv}B_f$ = bytes the collapse removes
1 vertex + the n_{uv} faces on the edge: 6 B + 2 x 2 B = 10 B (interior)
boundary edge (n_{uv} = 1): 8 B
score = 0.04 / 10 = 0.004
λ = 6.2e-3 : 0.004 < λ -> collapse
λ = 1.85e-3: 0.004 > λ -> keep
per round: no vertex in two collapses, cheapest first, <= 15 % of V,
reject if a surviving face would flip

Two independent byte prices: B_a = 6 B attributes of one vertex (position residual, DC colour, SH indices); B_f = 2 B connectivity of one face. Numbers are illustrative.